



Multidrug-resistant extended-spectrum beta-lactamase-producing *Escherichia coli* in poultry and cattle farms in India: Prevalence and survey-based risk factors

Hina Malik^a, Randhir Singh^{a,*}, Simranpreet Kaur^a, Neha Parmar^a, Anuj Tyagi^b, R.S. Aulakh^a, Jatinder Paul Singh Gill^a

^a Centre for One Health, Guru Angad Dev Veterinary and Animal Sciences University, Ludhiana, Punjab, India

^b College of Fisheries Sciences, Guru Angad Dev Veterinary and Animal Sciences University, Ludhiana, Punjab, India

ARTICLE INFO

Article history:

Received 17 March 2025

Revised 1 July 2025

Accepted 16 July 2025

Available online 25 July 2025

Editor: Dr Jon Hobman

Keywords:

ESBL *E. coli*

Poultry

Cattle

Antimicrobial resistance

Beta-lactam resistance genes

Cluster analysis

ABSTRACT

Objective: Unregulated use of antibiotics in animal-production leads to the emergence of multidrug-resistant (MDR) bacteria, including extended-spectrum beta-lactamase-producing *Escherichia coli* (ESBL *E. coli*), posing significant public health threats. This study aimed to assess and compare the role of cattle and poultry, along with farmworkers, in the emergence and spread of ESBL *E. coli* using a human-animal interface approach in One-Health framework.

Methods: Twenty cattle and 20 poultry farms in Ludhiana, India, were surveyed to determine antibiotic usage and biosecurity in farms. Animal faeces/poultry droppings, farmworkers hand swabs, and stool samples were examined for the presence of ESBL *E. coli*. The isolates were screened for antibiotic susceptibility and antimicrobial resistance genes. Generalized linear mixed models explored the variations in ESBL *E. coli* prevalence based on farm animal species, practices, and biosecurity measures.

Results: A higher prevalence of ESBL *E. coli* was identified in poultry (63.5%) compared to cattle (46%) and farmworkers (12%), which aligns with higher antibiotic use in poultry (85%) than cattle (70%). Most isolates (86.52%) were MDR, with poultry farm isolates having 97% MDR and 90.7% carrying the *bla*_{CTXM} gene. Generalized linear mixed models analysis revealed improved biosecurity and low antibiotic use were associated with lower ESBL *E. coli* prevalence.

Conclusions: This study highlights cattle and poultry farms as reservoirs for MDR and ESBL *E. coli*, contributing to rising environmental burden of antimicrobial resistance genes. It underscores the need to reduce antibiotic use in animal farming, particularly poultry, to address emerging antibiotic resistance and protect both animal and human health.

© 2025 The Author(s). Published by Elsevier Ltd on behalf of International Society for Antimicrobial Chemotherapy. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

1. Introduction

The emergence of beta-lactam antibiotic resistance in Enterobacteriaceae is a major One Health concern [1]. The production of extended-spectrum beta-lactamases (ESBLs) promotes the hydrolysis of broad-spectrum beta-lactam antibiotics such as cephalosporins and penicillin. ESBL-producing *Escherichia coli* (ESBL *E. coli*) poses substantial health risks not only in hospitals but also in communities due to its multidrug-resistance (MDR) and

its capacity to transmit resistant genes [2,3], with far-reaching implications for human, animal, and environmental health [4]. Widespread use of antibiotics in animal husbandry accelerates the emergence of resistance in human pathogens, further intensifying this global public health challenge [5].

Our previous review highlighted the significant use of critically important antibiotics in food-producing animals, contributing to antimicrobial resistance (AMR) and posing risks through contaminated food [6]. In this study, earlier findings were expanded upon to examine the specific risks posed by ESBL *E. coli* in livestock, poultry, and farmworkers, where data remain scarce. Given that India is recognized as a critical hotspot for AMR [7], a human-animal interface approach within One Health framework is essential to assess the distribution of ESBL *E. coli* across animals and farm-

* Corresponding author. Mailing address: Centre for One Health, Guru Angad Dev Veterinary and Animal Sciences University, Ferozpur road, Ludhiana, Punjab 141004, India.

E-mail address: sainirandhir74@gmail.com (R. Singh).

associated humans [8]. Thus, the aim of the study was to (1) assess on-farm antibiotic use, prevalent diseases, and biosecurity practices in cattle and poultry farms through questionnaires and surveys, (2) isolate and characterise ESBL and other beta-lactamase-producing *E. coli* strains from farm animals and workers, (3) compare resistance profiles across cattle, poultry and farmworkers, and (4) evaluate the association between the prevalence of ESBL *E. coli* with farm management practices (biosecurity and antibiotic use patterns) identified through surveys. These insights are crucial for developing targeted strategies to mitigate AMR and safeguard public health.

2. Materials and methods

2.1. Survey

The study was conducted in Ludhiana district of Punjab, a leading state in livestock and poultry production in India. A concise questionnaire was designed to gather data from farm owners on prevalent diseases, commonly used antibiotics, and biosecurity measures, including demographics (sex, age, education), farm type (large >50 cattle/5000 poultry or small), production method (intensive/semi-intensive/extensive) (Supplementary Data). During on-site evaluations, antibiotic storage and disposal were assessed, along with internal (disease control, sanitation) and external (infrastructure, location, feed/water sources, waste removal, vermin control) biosecurity measures. Following Kouam et al. [9], each biosecurity element was scored as either 1 or 0, and the total scores were subsequently converted to percentages.

2.2. Sample collection

Samples from cattle, poultry, and farmworkers were collected across seven tehsils of Ludhiana, Punjab, after obtaining ethical approval (DMCH/R&D/2021/7). Farm selection was based on the distribution of animal populations and the type of farm settings, ensuring a broad representation of both large and small-scale farms, and to have a balanced design. To minimise contamination, faecal samples were collected immediately after defecation. Human samples (stool/hand swabs) were voluntarily provided by farm workers after signing consent. A total of 488 samples were collected: 248 from cattle farms (11 large, 9 small farms), including farmworker stool ($n = 15$), hand swabs ($n = 33$), and cattle faeces ($n = 200$); and 240 from broiler farms (16 large, 4 small farms), including farmworker stool ($n = 6$), hand swabs ($n = 34$), and poultry droppings ($n = 200$).

2.3. Isolation and confirmation of ESBL-producing *E. coli*

Faecal samples were diluted in 10% buffered peptone water (1:9) and inoculated on HiChrome ESBL agar, followed by incubation at 37°C for 24–48 h. Presumptive ESBL *E. coli* was sub-cultured on eosin methylene blue agar, with species identification performed by Matrix Assisted Laser Desorption Ionization-Time of Flight Mass Spectrometry (MALDI-TOF MS). To confirm ESBL production, a modified Clinical and Laboratory Standards Institute combined disk test was performed using antibiotic disks: cefotaxime (CTX, 30 mcg), ceftazidime (CAZ, 30 mcg), cefotaxime-clavulanic acid (CEC, 30/10 mcg), ceftazidime-clavulanic acid (CAC, 30/10 mcg), in combination with phenylboronic acid (400 µg) and EDTA (292 µg) [10]. Isolates were considered ESBL-positive if the inhibition zone difference between CEC/CTX and/or CAC/CAZ exceeded 5 mm.

2.4. AmpC and carbapenemase (*Klebsiella pneumoniae* carbapenemase [KPC], metallo-beta-lactamase [MBL]) production test

To detect AmpC production, *E. coli* ATCC 25922 was inoculated on Mueller-Hinton Agar, and AmpC disks (50:50 mix of 100X tris-EDTA and normal saline) impregnated with test culture were placed adjacent to a cefoxitin (FOX, 30 mcg) disk. Indentation or flattening of the cefoxitin zone indicated AmpC production [11].

Carbapenemase production was tested by incubating meropenem (MRP, 10 mcg) disks with test culture broth, followed by placing the disks on Mueller-Hinton Agar plate inoculated with *E. coli* ATCC 25922. Reduced inhibition zones indicated carbapenemase production [12]. To determine carbapenemase type, further testing was conducted with meropenem disks in combination with phenylboronic acid (400 µg) and EDTA (292 µg) (see Supplementary Material).

2.5. Antibiotic susceptibility test

Confirmed ESBL-producing *E. coli* isolates were tested for antibiotic susceptibility using 16 antibiotics from 8 classes, based on the World Health Organization's (WHO) critically important antibiotics classification [13]. The Kirby-Bauer disk diffusion method was performed [14], following Clinical and Laboratory Standards Institute guidelines [15]. A complete list of antibiotics tested is available in the Supplementary Table 1. Multidrug resistance was determined according to Centers for Disease Control and Prevention (CDC) guidelines [16], and the Multiple Antibiotic Resistance (MAR) index was calculated [17].

2.6. DNA isolation and detection of selected antimicrobial resistance genes (ARGs)

Heat treatment method was used for the DNA extraction [18], and its quality and quantity were assessed using a NanoDrop spectrophotometer. PCR was performed to confirm 15 ARGs using specific primer pairs and amplification parameters (Supplementary Table 2). PCR (10 µL) reactions were prepared by mixing 2.5 µL PCR master mix, 0.3 µL (5 pmol) of each primer, 2.5 µL (10 ng) DNA template, and the volume was made up by adding nuclease-free water. Amplified products were analysed on 1.5% agarose gel with ethidium bromide and visualized using a gel documentation system.

2.7. Data analysis

Statistical analyses were conducted in R (version 4.2.0). Generalized linear mixed models (GLMMs) were built using a binomial distribution with a logit function to assess inter-individual variability in ESBL *E. coli* occurrence. The model included variables, i.e. farm animal species (cattle; $n = 240$, poultry; $n = 240$), farm type (large; $n = 322$, small; $n = 158$), and sample source (faeces; $n = 400$, farmworker hand swabs; $n = 67$, farmworker stool; $n = 13$), along with additional variables such as biosecurity score, owner's education, antibiotic presence at the farm, and treatment prescription, with farm identity was nested within area (tehsil) as a random variable. Akaike's information criterion corrected for small samples (AICc) was used for model selection, and both conditional coefficients of determination ($R^2_{GLMM(c)}$) and marginal coefficients ($R^2_{GLMM(m)}$) were reported, indicating the variance explained by the best model and ΔAIC_c . Conditional parameter estimates (β), odds ratio (OR), and confidence intervals (95% CI) were calculated using model averaging with a cumulative AICc of 95% [19]. The relative importance of each predictor variable was

Table 1
Farm owner demographics, antibiotic practices, and common disease patterns: survey overview.

Variables	Response	Studied farms		
		Cattle (n = 20)	Poultry (n = 20)	Total (n = 40)
Gender	Male	17 (85.00%)	20 (100.00%)	37 (92.50%)
	Female	3 (15.00%)	0 (0%)	3 (7.50%)
Age	30–45	9 (45.00%)	13 (65.00%)	22 (55.00%)
	46–60	5 (25.00%)	7 (35.00%)	12 (30.00%)
	>60	6 (30.00%)	0 (0.00%)	6 (15.00%)
Education	Graduate	10 (50.00%)	18 (90.00%)	28 (70.00%)
	Undergraduate	10 (50.00%)	2 (10.00%)	12 (30.00%)
Income share	Major	14 (70.00%)	16 (80.00%)	30 (75.00%)
	Medium to minor	6 (30.00%)	4 (20.00%)	10 (25.00%)
Farming	Contract	1 (5.00%)	14 (70.00%)	15 (37.50%)
	Self-owned	19 (95.00%)	6 (30.00%)	25 (62.50%)
Treatment prescription	Veterinarian	10 (50.00%)	15 (75.00%)	25 (62.50%)
	Non-veterinarian	10 (50%)	5 (25.00%)	15 (37.50%)
Antibiotics use/stored	Yes	14 (70.00%)	17 (85.00%)	31 (77.50%)
	No	6 (30.00%)	3 (15.00%)	9 (22.50%)
Common diseases	Mastitis	17 (85.00%)	NA	17 (85.00%)
	Reproductive disorders	16 (80.00%)	NA	16 (80.00%)
	Pyrexia	15 (75.00%)	NA	15 (75.00%)
	Diarrhoea	8 (40.00%)	3 (15.00%)	11 (27.50%)
	Respiratory disorder	6 (30.00%)	18 (90.00%)	24 (60.00%)
Commonly stored antibiotics	Gumboro	NA	7 (35.00%)	NA
	Quinolones	13 (65.00%)	15 (75.00%)	28 (70.00%)
	Aminoglycosides	7 (35.00%)	12 (60.00%)	19 (47.50%)
	Penicillin	8 (40.00%)	0 (0.00%)	8 (20.00%)
	3rd gen cephalosporin	9 (45.00%)	0 (0.00%)	9 (22.50%)
	Tetracycline	7 (35.00%)	12 (60.00%)	19 (47.50%)
	2nd gen cephalosporin	0 (0.00%)	2 (10.00%)	2 (5.00%)
	Polymyxin	1 (5.00%)	0 (0.00%)	1 (2.50%)
	Sulphonamide	2 (10.00%)	0 (0.00%)	2 (5.00%)

calculated by adding the AIC weights of each informative model in which the given variable was included. Tukey's General Linear Hypotheses Test was performed to compare average effects between groups when overall multivariable model significance was observed. Additionally, a stratified analysis of cattle and poultry farms was performed using similar GLMMs. To assess the correlation between phenotypic and genotypic antibiograms, a correlation matrix was plotted using the 'corrplot' version 0.84 [20]. Principal component analysis was performed on sensitivity zones of all tested antibiotics of the isolates (*FactoMineR*-version 4.2.1, [21]), followed by hierarchical clustering on principal components (*FactoMineR*-version 4.2.1, [22]). To visualize genotypic antibiotic resistance patterns, multiple correspondence analysis (MCA) was performed based on 13 of the 15 tested genes, excluding *qnrA* and *bla_{DHA}* due to only one positive isolate for each gene (*FactoMineR*-version 4.2.1) [23].

3. Results

A survey was conducted to identify antibiotic usage, prevalent diseases, and biosecurity measures in farms, and collected samples from animals and farmworkers to evaluate ESBL *E. coli* occurrence and resistance patterns. The results revealed significant antibiotic usage and varying levels of treatment and biosecurity measures among the farms studied. Antibiotic use was widespread, with 77.5% of farms storing them for later use. Poultry farms had a higher usage rate (85%) compared to cattle farms (70%). Commonly stored antibiotic classes included quinolones (70%), aminoglycosides (47.5%), and tetracycline (47.5%), with poultry farms exhibiting greater use of these classes. Diseases were prevalent in both farm types, with mastitis and reproductive disorders common in cattle, and chronic respiratory disease and Gumboro frequently reported in poultry. Only 62.5% of farms sought treatment from veterinarians, while 37.5% relied on non-veterinarians. Table 1 provides a comprehensive summary of survey responses, including demographic details and antibiotic usage patterns.

Table 2
Distribution of ESBL *Escherichia coli* isolates in cattle and poultry farm samples.

Source	Cattle	Poultry
Animal faeces	92/200 (46%)	127/200 (63.5%)
Farmworker hand swabs	6/33 (18.18%)	2/34 (5.88%)
Farmworker stool	3/15 (20%)	0/6 (0%)
Grand Total	101/248 (40.72%)	129/240 (53.75%)

3.1. Occurrence of ESBL-producing *E. coli* (ESBL *E. coli*)

A total of 230 ESBL *E. coli* were detected in 488 samples from both cattle and poultry farms, with a higher occurrence in poultry (Table 2). The GLMM analysis found that farm animal species, sample source, and biosecurity score as significant predictors of ESBL *E. coli* occurrence in the best-fit model ($\Delta AIC_C = 67.77$, $R^2_{GLMM(m)} = 0.241$, $R^2_{GLMM(c)} = 0.296$). The analysis identified biosecurity score, farm animal species, and sample source as the key predictors of ESBL *E. coli* occurrence ($\Sigma AIC_{C\phi} = 1$), followed by treatment prescription ($\Sigma AIC_{C\phi} = 0.43$). Farm type, owner education, and antibiotic presence showed minimal impact. Farms with higher biosecurity scores had an OR of 0.93 (95% CI: 0.90–0.96) for ESBL *E. coli* occurrence (Supplementary Fig. 1). Poultry had an OR of 3.48 (95% CI: 1.79–6.79) compared to cattle (Fig. 1A). ESBL *E. coli* was more prevalent in animal faeces than in farmworker hand swabs (OR: 0.07, 95% CI: 0.03–0.17) or stool (OR: 0.25, 95% CI: 0.06–1.06) (Fig. 1B). Detailed OR and CI for all variables can be found in Supplementary Table 3.

3.2. ESBL *E. coli* in cattle farms

Within cattle farms, the best-fit model was indicated by a ΔAIC_C of 81.07, with $R^2_{GLMM(m)} = 0.783$ and $R^2_{GLMM(c)} = 0.783$. In averaged model, strong support was found for the influence of farm type, sample source, biosecurity score, antibiotic presence

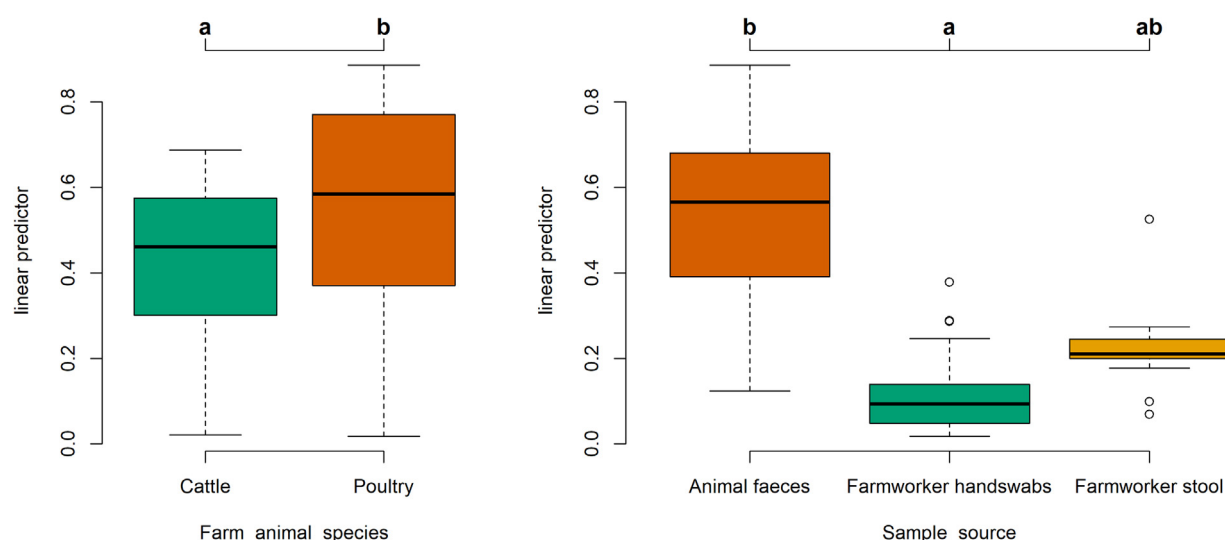


Fig. 1. Comparison of ESBL *Escherichia coli* occurrence between farm animal species (A) and sample source (B).

($\Sigma AIC_C = 1$), and treatment by professionals ($\Sigma AIC_C = 0.92$) on ESBL *E. coli* occurrence, while moderate support was noted for owner education status ($\Sigma AIC_C = 0.50$). Large cattle farms had a higher occurrence of ESBL *E. coli* (50.78%) than small farms (30%) ($P < .01$). Farms that used or stored antibiotics were much more likely to have high ESBL *E. coli* levels (OR: 9.00; CI: 2.64–306.7, $P < .05$). Farms that received treatment from veterinarians demonstrated a lower likelihood of ESBL *E. coli* prevalence (OR: 0.20; CI: 0.04–0.93, $P < .05$). OR and CI for all variables are summarized in Supplementary Table 4.

3.3. ESBL *E. coli* in poultry farms

Within poultry farms, the best-fit model was indicated by a ΔAIC_C of 7.81, with $R^2_{GLMM(m)} = 0.087$ and $R^2_{GLMM(c)} = 0.164$. Averaged model results showed strong support for the influence of sample source ($\Sigma AIC_C = 1$) and farm type ($\Sigma AIC_C = 0.69$) on the occurrence of ESBL *E. coli*, similar to the omnibus model. Higher ESBL *E. coli* occurrence was noted in samples from large farm compared to small farms, and in poultry samples compared to farmworker samples. Intriguingly, the other investigated predictor variables, namely biosecurity score, antibiotic usage/storage, and treatment prescription, exhibited weak support on ESBL *E. coli* occurrence in poultry specific model. OR and CI can be found in Supplementary Table 5.

3.4. Antibiotic sensitivity pattern and beta-lactamase production in ESBL *E. coli* isolates

ESBL *E. coli* isolates from cattle and poultry showed significant resistance to multiple antibiotic classes, particularly third-generation cephalosporins, quinolones, and aminoglycosides (Fig. 2A, Supplementary Table 6). Cattle isolates exhibited significant resistance to beta-lactams, tetracycline, and amikacin, whereas poultry isolates demonstrated high resistance to cephalosporins and quinolones. Of the isolates, 86.52% of the isolates were multidrug resistant, with higher proportions observed in poultry (96.90%) compared to cattle (73.27%). In cattle farm isolates, 72.83% of animal faecal isolates, 100% of farmworker stool isolates, and 66.67% of hand swabs exhibited MDR. MDR was detected in 96.85% of poultry droppings and 100% of farmworker hand swabs samples from poultry farms. A total of 36 distinct MDR profiles were observed, with the most prevalent (13%) showing resistance to beta-lactams, quinolones, aminoglycosides, and tetra-

cycline. Additionally, 93.91% of isolates had a MAR index greater than 0.2, including 100% of poultry farm isolates and 86.14% of cattle farm isolates. Farmworker stool isolates exhibited the highest MAR indices, followed by animal faeces and farmworker hand swab samples. Detailed values are summarized in Supplementary Table 7. Among the 230 ESBL *E. coli* isolates, additional beta-lactamase production was identified, with 90 isolates producing AmpC, 1 producing KPC, and 2 producing both KPC and MBL enzymes (Supplementary Fig. 2). Notably, 43.41% of the ESBL-AmpC co-producers were from poultry farms, while 33.66% originated from cattle farms.

3.5. ARGs in *E. coli* isolates

All 230 ESBL *E. coli* isolates were examined for 15 ARGs, including beta-lactam, aminoglycoside, tetracycline, quinolone, and sulphonamide resistance genes (detailed in Supplementary Table 8, Fig. 2B). The most prevalent gene was *bla*_{AmpC}, detected in 83.48% of isolates, followed by *bla*_{CTXM} (80.87%) and *bla*_{TEM} (57.39%). Poultry isolates exhibited a higher overall occurrence of ARGs compared to cattle, with notable differences in gene carriage. For instance, *bla*_{CTXM} was more prevalent in poultry isolates (91.34% vs. 66.30%), while *bla*_{AmpC} was predominant in cattle isolates (96.74% vs. 75.59%).

3.6. Association between AMR phenotype and resistance genes

Several ARGs showed significant correlation ($P \leq .05$) with phenotypic AMR, as illustrated in Fig. 3. Strong positive correlations were observed between the *bla*_{CTXM} gene and resistance to ceftazidime ($r = 0.9$), ampicillin ($r = 0.9$), and cefotaxime ($r = 0.7$), with moderate correlations noted for ceftazidime ($r = 0.4$) and a negative correlation with *bla*_{SHV} ($r = -0.5$). Hierarchical clustering on principal component analysis based on phenotypic resistance revealed three distinct clusters: the first ($n = 38$) was dominated by poultry isolates highly resistant to cephalosporins and quinolones; the second ($n = 164$) included mostly cattle and poultry isolates with high resistance to cephalosporin, quinolones, and tetracycline, but lower resistance to aminoglycosides; and the third ($n = 28$) consisted solely of cattle and cattle farmworkers isolates showing lesser overall resistance (Fig. 4A).

MCA of ARG profiles demonstrated overlapping clusters corresponding to their sources of isolation. Dimension 1 was significantly influenced by the presence of *tet(A)* and *bla*_{CTXM15}, while Di-

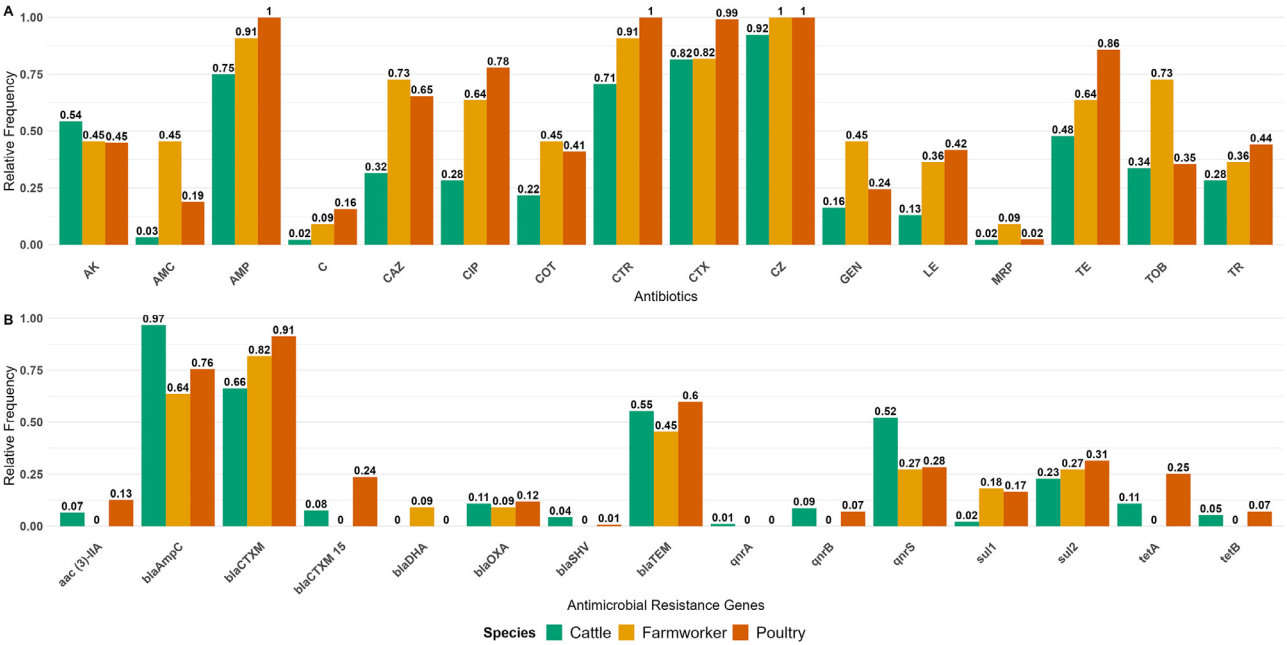


Fig. 2. Comparison of antibiotic resistance patterns (A) and AMR gene distribution (B) in ESBL-producing *Escherichia coli* from cattle, poultry, and farmworkers. Proportions are displayed as labelled text. Antibiotic abbreviations: AK, amikacin; AMC, amoxicillin-clavulanic acid; AMP, ampicillin; C, chloramphenicol; CAZ, ceftazidime; CIP, ciprofloxacin; COT, co-trimoxazole; CTR, ceftriaxone; CTX, cefotaxime; CZ, ceftazolin; GEN, gentamicin; LE, levofloxacin; MRP, meropenem; TE, tetracycline; TOB, tobramycin; TR, trimethoprim.

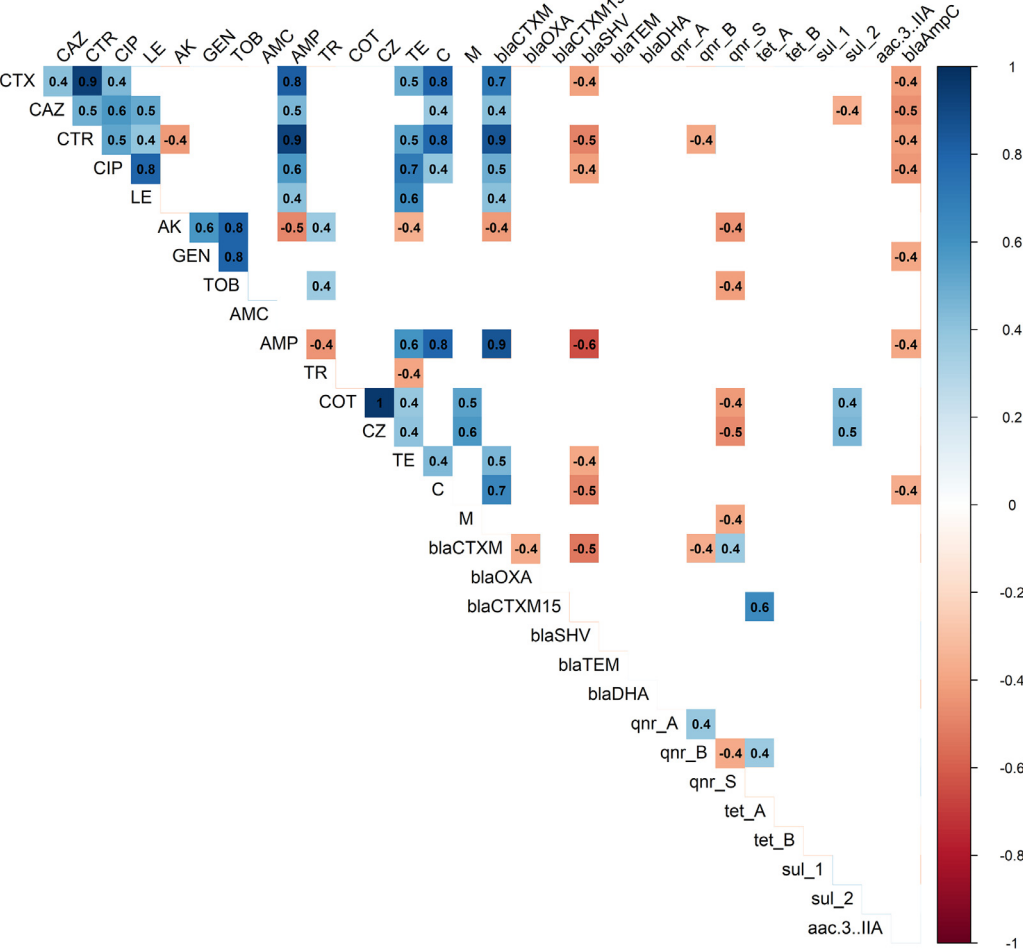


Fig. 3. Correlation matrix showing significant correlation among phenotypic and genotypic resistance profiles along with r values.

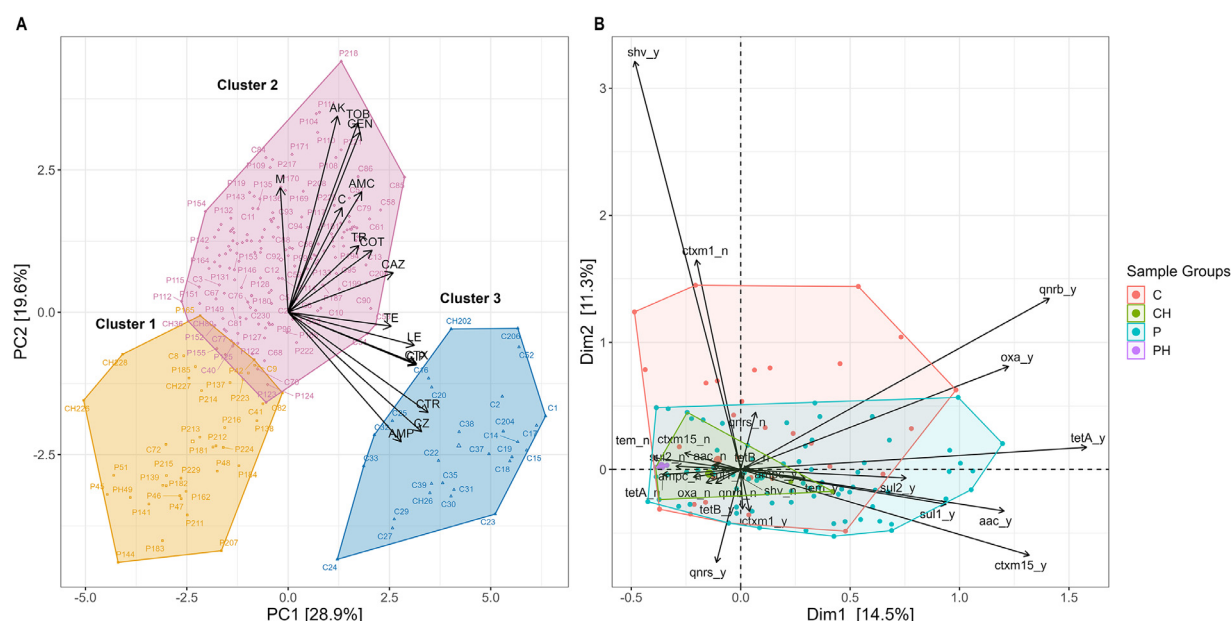


Fig. 4. Hierarchical clustering on principal component analysis (HCPC) (A) and multiple correspondence analysis (MCA) (B) of ESBL *Escherichia coli* isolates showing clustering patterns by source. Source labels: Cattle [C], cattle farmworker [CH], poultry [P], poultry farmworker [PH].

mension 2 was characterized by the absence of *tet(A)* and *bla_{CTXM}*, and the presence of *bla_{SHV}* (Fig. 4B). The observed clustering was largely driven by a higher prevalence of ARGs such as *tet(A)* and *bla_{CTXM15}* in poultry and cattle isolates, with this cluster primarily spread along Dimension 1. In contrast, a smaller cluster, comprising a few cattle isolates and distributed along Dimension 2, exhibited the presence of *bla_{SHV}* gene and/or absence of the *bla_{CTXM}* gene.

Overall, a strong correlation was identified between phenotypic and genotypic resistance patterns, indicating that the clustering of isolates mostly reflects their sources of isolation.

4. Discussion

The emergence of ESBL *E. coli* has raised significant concerns, particularly due to its resistance to third-generation cephalosporins, leading to the prevalence of drug-resistant infections. This study investigates the prevalence of ESBL *E. coli*, AMR patterns, and associated resistance genes among farm animals and their handlers in Punjab, India, employing a human-animal-focused approach within One Health framework. The survey of this study showed widespread antibiotic use in both cattle and poultry farming, with multiple antibiotic classes frequently used and stored, raising concerns about the potential for increased resistance. The survey results indicated that promoting good hygiene practices can prevent disease spread, reducing antibiotic use and financial burden on farmers, many of whom depend entirely on animal farming for their income. Most farmers were educated, yet many turned to non-veterinarians for animal treatment, which could negatively impact animal health and potentially increase antibiotic use. The study showed that all examined cattle and poultry farms were positive for ESBL *E. coli*, with overall prevalence rates of 40.72% in cattle and 53.75% in poultry, indicating these farms may serve as reservoirs for ESBL *E. coli*.

Several studies documented ESBL *E. coli* prevalence in Indian cattle and poultry farms, showing lower rates in cattle (4.35%) [24] and generally higher rates in poultry (6–87%) [25,26]. Poultry worldwide has higher ESBL rates than cattle, with variations across countries [27–29], partly due to higher antibiotic use in poultry production [5]. Data from this study indicated that 85% of poultry

farms used antibiotics for disease prevention/growth promotion, a significant increase from the 67% reported in a previous study from Punjab [25]. The increase in prevalence from 42% [25] to 53.75% in this study highlights the escalating AMR problem in the poultry production sector in Punjab.

A higher prevalence of ESBL *E. coli* was found in larger cattle farms, mirroring results from poultry farming. This indicated that intensive farming carries a greater risk compared to small-scale operations [30]. This association may arise from higher animal density and more antibiotic use for disease management [31]. Cattle farms that stored antibiotics for later use and sought treatment from non-veterinarians were more likely to have higher levels of ESBL *E. coli*. These practices exacerbate the AMR problem in India [32], primarily due to inadequate regulations on antibiotic sales and limited access to veterinarians.

Interestingly, data from poultry farms showed that biosecurity score, antibiotic usage/storage, and treatment prescription did not significantly affect ESBL *E. coli* occurrence. This is likely because 85% of farms use antibiotics for disease prevention or growth promotion, which limits the ability to statistically evaluate other contributing factors. The high prevalence of MDR ESBL *E. coli* in poultry droppings can be attributed to the intense selection pressure resulting from antibiotic use. Many of these isolates also produce other beta-lactamase enzymes, including AmpC, KPC, and MBL, which severely limits treatment options. The plasmid-mediated nature of these resistance genes facilitates their spread, creating major public health challenges [33]. To reduce the spread of resistant bacteria in poultry farms, it is crucial to prioritize the judicious use of critically important antibiotics, complemented with improved environmental management, regular monitoring, and increased awareness of responsible antibiotic use among farmers.

This study revealed a high prevalence (93.91%) of MDR, particularly among poultry isolates (100%). Previous studies reported varying MDR rates of 30% to 89% in ESBL *E. coli* from poultry [34,35]. Notably, Waade et al. [36] reported lower MDR (53%) in cattle, suggesting a concerning trend in poultry. A positive correlation was observed between resistant phenotypes across antibiotic classes in ESBL *E. coli* isolates, aligning with other studies [37]. Nearly all ESBL *E. coli* isolates (99.13%) were resistant to at least

one antibiotic class besides beta-lactams, predominantly tetracycline and quinolones. Findings of this study show that tetracycline, cephalosporins, and quinolones were the most commonly used antibiotics on these farms, supporting this resistance trend. The high prevalence of MDR is likely due to cross-resistance mechanisms and sharing of ARGs among microbial communities.

Three distinct clusters of phenotypic resistance were observed in ESBL *E. coli* isolates: one dominated by poultry isolates, another comprising mixed cattle, poultry, and poultry farm workers isolates, and a third group consisting of cattle isolates and cattle farm workers. These clusters likely reflect the varying antibiotic usage practices in cattle and poultry farming, which influences the resistance profiles. The MCA demonstrated how specific ARGs influenced these patterns and contributed to segregation of cattle and poultry isolates. Notably, most cattle isolates carrying the *bla_{SHV}* gene exhibited sensitivity to third-generation cephalosporins, while *bla_{CTXM}* was more prevalent in poultry isolates, correlating with their resistance to cefotaxime, aligning with previous studies [37,38]. A moderate positive correlation was found between the *tet(A)* gene and *bla_{CTXM15}*, indicating potential co-selection mechanisms.

Farmworkers, working closely with cattle and poultry, are at risk of acquiring resistant microorganisms. This study found that 12% of farmworkers tested positive for ESBL *E. coli*. Notably, while poultry had a higher prevalence overall, poultry farm workers exhibited a lower rate (5%) compared to cattle farm workers (18.75%). This difference is likely due to cattle farmworkers having more frequent contact with animals during feeding, milking, and cleaning. However, the investigation faced limitations due to farmworkers' reluctance to provide stool samples. Previous studies show positive ESBL *E. coli* cases among farmworkers, with Nossair et al. [37] documenting a 20% prevalence among livestock and poultry handlers. Furthermore, 9% of farmworker isolates exhibited resistance to meropenem, consistent with findings by Nossair et al. [37]. Factors contributing to the presence of ESBL *E. coli* in farmworkers may include inadequate hygiene standards, antibiotic usage, and close animal-human contact. While this study demonstrated similar resistance patterns between animal and farmworker isolates, genomic analysis would be required to definitively establish transmission routes between animals and farm workers.

The findings regarding the extensive antibiotic use in cattle and poultry farms of Ludhiana, Punjab, reflect similar AMR patterns observed across Southeast Asia, as highlighted in a previous review [6]. This reinforces the urgent need for tailored strategies to effectively combat AMR within different farming systems. Although the human-animal-focused approach provided essential baseline data, future studies should expand to include environmental components to achieve comprehensive One Health surveillance.

5. Conclusion

This study underscores that cattle and poultry farms serve as reservoirs for MDR and ESBL *E. coli*, with significantly higher occurrences of ESBL *E. coli* in poultry. The alarming levels of MDR and burden of ARGs raise pressing concerns regarding the limited availability of effective treatment. This study highlights the crucial need for implementing tailored strategies, including stringent regulations on antibiotic usage and improved waste management in poultry farms, while enhancing biosecurity measures and ensuring that antibiotics are prescribed by veterinarians. These customized approaches are essential for addressing the specific challenges of AMR in these farming systems, ultimately safeguarding animal welfare and human health.

Acknowledgements: We are grateful to Wasimuddin, Senior Scientist, Norwegian Veterinary Institute, for his scientific advice and all farmers for their constant help and support during our field work.

Funding: This research was funded by Indian Council of Agriculture Research project, 'Niche area of excellence (NAE) programme entitled Antimicrobial resistance: Animal-Human interface'.

Ethical approval: The study was reviewed and approved by the Ethics Committee of Dayanand Medical College and Hospital, under the reference no: DMCH/R&D/2021/7, dated 20/01/2021.

Author contributions: HM and RS conceived the study. HM carried out laboratory experiments, statistical analyses, and wrote the manuscript. NP contributed in the sample collection, and RS edited the manuscript. All authors contributed to the article and approved the submitted version.

Declaration of competing interests: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jgar.2025.07.013.

References

- [1] Wieler LH. "One health"—linking human, animal and environmental health. *Int J Med Microbiol* 2014;304:775–6. doi:10.1016/j.ijmm.2014.08.014.
- [2] Peirano G, Pitout JDD. Extended-spectrum β -lactamase-producing Enterobacteriaceae: update on molecular epidemiology and treatment options. *Drugs* 2019;79:1529–41. doi:10.1007/s40265-019-01180-3.
- [3] McInnes RS, McCallum GE, Lamberte LE, van Schaik W. Horizontal transfer of antibiotic resistance genes in the human gut microbiome. *Curr Opin Microbiol* 2020;53:35–43. doi:10.1016/j.mib.2020.02.002.
- [4] Bevan ER, Jones AM, Hawkey PM. Global epidemiology of CTX-M β -lactamases: temporal and geographical shifts in genotype. *J Antimicrob Chemother* 2017;72:2145–55. doi:10.1093/jac/dkx146.
- [5] Van Boeckel TP, Brower C, Gilbert M, Grenfell BT, Levin SA, Robinson TP, et al. Global trends in antimicrobial use in food animals. *Proc Natl Acad Sci USA* 2015;112:5649–54. doi:10.1073/pnas.1503141112.
- [6] Malik H, Singh R, Kaur S, Dhaka P, Bedi JS, Gill JPS, et al. Review of antibiotic use and resistance in food animal production in WHO South-East Asia Region. *J Infect Public Health* 2023;16:172–82. doi:10.1016/j.jiph.2023.11.002.
- [7] Zhao C, Wang Y, Mulchandani R, Van Boeckel TP. Global surveillance of antimicrobial resistance in food animals using priority drugs maps. *Nat Commun* 2024;15:763. doi:10.1038/s41467-024-45111-7.
- [8] Tseng CH, Liu CW, Liu PY. Extended-spectrum β -lactamases (ESBL) producing bacteria in animals. *Antibiotics* 2023;12:661. doi:10.3390/antibiotics12040661.
- [9] Kouam MK, Jacouba M, Nsangou IN, Teguia A. Assessment of biosecurity level in small-scale broiler farms in the western highlands of Cameroon (Central Africa). *Trop Anim Health Prod* 2018;50:1529–38. doi:10.1007/s11250-018-1591-x.
- [10] Poulou A, Grivakou E, Vroni G, Koumaki V, Pittaras T, Pournaras S, et al. Modified CLSI extended-spectrum β -lactamase (ESBL) confirmatory test for phenotypic detection of ESBLs among Enterobacteriaceae producing various β -lactamases. *J Clin Microbiol* 2014;52:1483–9. doi:10.1128/JCM.03361-13.
- [11] Black J, Moland ES, Thomson K. AmpC disk test for detection of plasmid-mediated AmpC β -lactamases in Enterobacteriaceae lacking chromosomal AmpC β -lactamases. *J Clin Microbiol* 2005;43:3110–13. doi:10.1128/JCM.43.7.3110-3113.2005.
- [12] van der Zwaluw K, de Haan A, Pluister GN, Bootsma HJ, de Neeling AJ, Schouls LM. The Carbapenem Inactivation Method (CIM), a simple and low-cost alternative for the Carba NP test to assess phenotypic carbapenemase activity in Gram-negative rods. *PLoS ONE* 2015;10:e0123690. doi:10.1371/journal.pone.0123690.
- [13] World Health Organization WHO list of critically important antimicrobials (CIA). Geneva, Switzerland; 2018.
- [14] Hudzicki J. Kirby-Bauer disk diffusion susceptibility test protocol author information. *Am Soc Microbiol* 2009;15:1–23.
- [15] CLSI M100: performance standards for antimicrobial susceptibility testing. 30th ed. Wayne, Pennsylvania, 19087, USA: Clinical and Laboratory standards Institute; 2020.

- [16] Magiorakos AP, Srinivasan A, Carey RB, Carmeli Y, Falagas ME, Giske CG, et al. Multidrug-resistant, extensively drug-resistant and pandrug-resistant bacteria: an international expert proposal for interim standard definitions for acquired resistance. *Clin Microbiol Infect* 2012;18:268–81. doi:10.1111/j.1469-0691.2011.03570.x.
- [17] Krupperman PH. Multiple antibiotic resistance indexing of *Escherichia coli* to identify high-risk sources of faecal contamination of water. *Appl Environ Microbiol* 1983;46:165–70. doi:10.1007/s11356-014-3887-3.
- [18] Dashti AA, Jadaon MM, Abdulsamad AM, Dashti HM. Heat treatment of bacteria: a simple method of DNA extraction for molecular techniques. *Kuwait Med J* 2009;41:117–22.
- [19] Burnham KP, Anderson DR. Model selection and multimodel inference: a practical information-theoretic approach. 2nd ed. New York, NY: Springer; 2002. doi:10.1007/b97636.
- [20] Wei T, Simko V. R package “corrplot”: visualization of a correlation matrix (Version 0.84) <https://CRAN.R-project.org/package=corrplot>.
- [21] Lê S, Josse J, Husson F. FactoMineR: an R package for multivariate analysis. *J Stat Softw* 2008;25:1–18. doi:10.18637/jss.v025.i01.
- [22] Husson F, Josse J, Pagès J. Principal component methods-hierarchical clustering-partitional clustering: why would we need to choose for visualizing data. *Appl Math Dep* 2010;17:1–17. http://factominer.free.fr/more/HPCP_husson_josse.pdf.
- [23] Husson F, Lê S, Pagès J. Exploratory multivariate analysis by example using R. Boca Raton: CRC Press; 2017. doi:10.1201/b21874.
- [24] Das L, Borah P, Sharma RK, Malakar D, Saikia GK, Sharma K, et al. Phenotypic and molecular characterization of extended spectrum β -lactamase producing *Escherichia coli* and *Klebsiella pneumoniae* isolates from various samples of animal origin from Assam, India. *BioRxiv* 2020:122705. doi:10.1101/2020.05.28.122705.
- [25] Brower CH, Mandal S, Hayer S, Sran M, Zehra A, Patel SJ, et al. The prevalence of extended-spectrum beta-lactamase-producing multidrug-resistant *Escherichia coli* in poultry chickens and variation according to farming practices in Punjab, India. *Environ Health Perspect* 2017;125:077015. doi:10.1289/EHP292.
- [26] Chowdhury M, Bardhan R, Pal S, Banerjee A, Batabyal K, Joardar SN, et al. Comparative occurrence of ESBL/AmpC beta-lactamase-producing *Escherichia coli* and *Salmonella* in contract farm and backyard broilers. *Lett Appl Microbiol* 2022;74:53–62. doi:10.1111/lam.13581.
- [27] Gelalcha BD, Ensermu DB, Agga GE, Vancuren M, Gillespie BE, D'Souza DH, et al. Prevalence of antimicrobial resistant and extended-spectrum beta-lactamase-producing *Escherichia coli* in dairy cattle farms in East Tennessee. *Foodborne Pathog Dis* 2022;19:408–16. doi:10.1089/fpd.2021.0101.
- [28] Liu Z, Wang K, Zhang Y, Xia L, Zhao L, Guo C, et al. High prevalence and diversity characteristics of *bla*_{NDM}, *mcr*, and *bla*_{ESBLs} harboring multidrug-resistant *Escherichia coli* from chicken, pig, and cattle in China. *Front Cell Infect Microbiol* 2022;11:755545. doi:10.3389/fcimb.2021.755545.
- [29] Weber LP, Dreyer S, Heppelmann M, Schauler K, Homeier-Bachmann T, Bachmann L. Prevalence and risk factors for ESBL/AmpC-E. *coli* in pre-weaned dairy calves on dairy farms in Germany. *Microorganisms* 2021;9:2135. doi:10.3390/microorganisms9102135.
- [30] Gālina D, Balins A, Valdovska A. The prevalence and characterization of fecal extended-spectrum-beta-lactamase-producing *Escherichia coli* isolated from pigs on farms of different sizes in Latvia. *Antibiotics* 2021;10:1099. doi:10.3390/antibiotics10091099.
- [31] Wang H, Qi JF, Qin R, Ding K, Graham DW, Zhu YG. Intensified livestock farming increases antibiotic resistance genotypes and phenotypes in animal feces. *Commun Earth Environ* 2023;4:123. doi:10.1038/s43247-023-00790-w.
- [32] Mutua F, Sharma G, Grace D, Bandyopadhyay S, Shome B, Lindahl J. A review of animal health and drug use practices in India, and their possible link to antimicrobial resistance. *Antimicrob Resist Infect Control* 2020;9:103. doi:10.1186/s13756-020-00760-3.
- [33] Bush K, Jacoby GA. Updated functional classification of β -lactamases. *Antimicrob Agents Chemother* 2010;54:969–76. doi:10.1128/AAC.01009-09.
- [34] De Koster S, Ringenier M, Lammens C, Stegeman A, Tobias T, Velkers F, et al. ESBL-producing, carbapenem- and ciprofloxacin-resistant *Escherichia coli* in Belgian and Dutch broiler and pig farms: a cross-sectional and cross-border study. *Antibiotics* 2021;10:945. doi:10.3390/antibiotics10080945.
- [35] Sonola VS, Katakweba AS, Misinzio G, Matee MIN. Occurrence of multi-drug-resistant *Escherichia coli* in chickens, humans, rodents and household soil in Karatu, Northern Tanzania. *Antibiotics* 2021;10:1137. doi:10.3390/antibiotics10091137.
- [36] Waade J, Seibt U, Honscha W, Rachidi F, Starke A, Speck S, et al. Multidrug-resistant enterobacteria in newborn dairy calves in Germany. *PLoS ONE* 2021;16:e0248291. doi:10.1371/journal.pone.0248291.
- [37] Nossair MA, Abd El Baqy FAA, Rizk MSY, Elaadli H, Mansour AM, El-Aziz AHA, et al. Prevalence and molecular characterization of extended-spectrum β -lactamases and AmpC β -lactamase-producing Enterobacteriaceae among human, cattle, and poultry. *Pathogens* 2022;11:852. doi:10.3390/pathogens11080852.
- [38] Ahmed S, Das T, Islam MZ, Herrero-Fresno A, Biswas PK, Olsen JE. High prevalence of *mcr*-1-encoded colistin resistance in commensal *Escherichia coli* from broiler chicken in Bangladesh. *Sci Rep* 2020;10:1–13. doi:10.1038/s41598-020-75608-2.